

LAITH ASSAF

East Lansing, MI | (248) 480-5783 | assaflai@msu.edu | [linkedin.com/in/laith-assaf-](https://www.linkedin.com/in/laith-assaf-) | github.com/Novapool

EDUCATION

Michigan State University, College of Engineering

East Lansing, MI

BS Computer Science, AI/ML | Dean's List x3

August 2022 - May 2027

- **Relevant Coursework:** Data Structures & Algorithms (CSE 331), Cloud Computing (CSE 380), Software Design (CSE 335), Computer Systems (CSE 325)

EXPERIENCE

PC Technician Intern

May 2026 - Present

Onsite Support Team, Delta Dental

Okemos, MI

- Deployed and configured **150+** workstations (monitors, docking stations, keyboards, mice) across a new corporate office buildout, imaging endpoints via **Ivanti** MDM to enforce standardized enterprise configuration
- Developing an automated ticket creation workflow in **Power Automate** to streamline IT ops; performed PC imaging and disk backups and drive kills across endpoint lifecycle tasks

IT and Media Services Support Associate

May 2023 - Present

MSU College of Osteopathic Medicine

East Lansing, MI

- Configured and maintained 100+ classroom technology systems (projectors, audio systems, video conferencing) across 4 locations supporting **1,200+ students**, resolving 50-70 technical tickets weekly with 5-10 minute average response time
- Engineered systematic troubleshooting framework maintaining **99.5% classroom uptime** and preventing \$50K annual losses through diagnostic methodology, resolving network connectivity, display output, audio routing, and Zoom conferencing issues within 5-minute SLA

PROJECTS

SoundSense - Real-Time Audio Classification | *YAMNet, WebSocket* | **MHacks Winner**

October 2025

- **Won at MHacks 2025** (300+ participants, 24-hour hackathon) building real-time audio classification system using **YAMNet** CNN on MemryX MXA2301 edge accelerators, achieving **95% accuracy** at <50ms latency across 521 AudioSet classes; delivered hazard alerts via **Flask** + **WebSocket** pipeline to a live web interface for hearing-impaired users

Bonfire Platform & Ember Framework | *TypeScript, Node.js, Socket.io, Next.js*

December 2025 - Present

- Designed and published **Ember**, an open-source TypeScript framework for multiplayer party games (@**bonfire-ember/core/server/client** npm packages), implementing phase state machine, WebSocket room orchestration, and player lifecycle into a 5-hook API enabling **80%+ code reuse** across new game builds
- Architected **multi-room orchestration system** on Express + **Socket.io** namespaces with **Firebase Realtime DB** state sync, hosting 4 concurrent games on a single **Railway** server with configurable disconnect strategies and TTL-based cleanup supporting 1,000+ concurrent rooms

LEADERSHIP

Michigan State University | AI Club

East Lansing, MI

Workshop Coordinator

January 2025 - Present

- Developed and delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), AI ethics, and ML fundamentals (**PyTorch**) to 500+ attendees across 15 sessions, recording 90% retention and 4.8/5 satisfaction score

Michigan State University | AI Club

East Lansing, MI

Project Lead

January 2025 - May 2025

- Led 4-student team building Academic Performance Predictor using **GPT-4 API** and prompt engineering, delivering **80% prediction accuracy** across 50+ courses with OpenAI API and agile sprints

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, TypeScript

AI/ML: PyTorch, OpenAI API, YAMNet, Ollama

Web & Mobile: FastAPI, Flask, React, Node.js, Express, Socket.io, REST APIs, WebSockets, Firebase

Infrastructure & Tools: Git, Linux, Docker, Railway, Vercel, Active Directory, Intune, Ivanti, Power Automate